The Great Orme touch-screen interactive tour

FINAL TEXT V4: MARCH 9 2013

TOP MENU: choose Welsh / English text

MAIN MENU PAGE:

You are standing near the summit of one of Wales' most spectacular Country Parks. Over half a million visitors come here each year to admire the beautiful scenery and catch a glimpse of the rare species of wildlife, both animals and plants, which can only be found here.

Use the maps to **EXPLORE** the Great Orme's many attractions, or join our Country Park Warden to **DISCOVER** some of the Great Orme's secrets…

TITLES

MAP
CHOOSE LANGUAGE
CHOOSE CATEGORY
GEOLOGY
HISTORY
WILDLIFE
THINGS TO DO
ATTRACTIONS
PLACES TO EAT

DISCOVER GEOLOGY

INTRO PAGE 1:

Discover how the Great Orme was formed

The Great Orme is a spectacular headland 3 km long and 207 metres high, formed from rock created over 300 million years ago and shaped by the forces of nature ever since. We call the study of rocks **geology**, and the Great Orme has many nationally important geological features: let's take a tour through those 300 million years…

Q1 Do you think that the Great Orme is the remains of an extinct volcano? □

Or the remains of millions of tiny sea creatures? □ ✔

PAGE 2: The Great Orme is made of limestone. All this area was once at the bottom of the sea, and the shells of all the sea creatures and corals were crushed together to form rock. You can see fossils of many of them in the limestone at Bishop's Quarry,

but please do not try to remove them. As limestone is very hard, rock was taken from here to build a palace for the Bishop of Bangor in the 13th Century. *(pic)*

PAGE 3: Volcanic forces under the Earth's crust made it shift and buckle, and the once-flat layers of rock were ripped apart in different directions and left lying at various angles, as you can see on the cliffs. *(pic)*

Q2: Molten rock and gasses were forced up through the rocks: what do you think this would do?

It created all the cracks and caves □

It deposited minerals and metals in the limestone □✔

PAGE 4: The cracks and caves were formed by later weathering; the volcanic magma created veins of copper minerals, including striking green Malachite. *(pic)*

Q3: Can you guess what had the biggest impact on shaping the Great Orme?

Ice □ ✔ Water □

PAGE 5: During the last Ice Age, there would have been over a kilometre of ice above our heads. The pressure of the ice in the glaciers scraped across the rock, creating the rounded shape we see today. *(pic)*

Q4: Do you think this stone was:

Dropped here by a glacier □✔ A man-made standing stone □

PAGE 6: As the glaciers moved, they ripped up large stones and dropped them again. These stones are called glacial erratics. This one is called "The Mother & Daughter" *pic*.

PAGE 7:
You can explore the map to see where all these features are, or discover many more fascinating features of the Great Orme's geology by **Taking a Tour** *(brochure pic) + pop-up text:*

The Summit Trails leaflet can be bought at the visitor centre shop, and contains three scenic walks with stunning views; running from the town of Llandudno to the summit of the Great Orme, or followed in reverse from the summit to Llandudno. The tracks are steep and uneven in places, and walking boots are recommended.

- Haulfre Gardens Trail: 1 mile. Allow about 1 hour.
- Happy Valley Trail: 1½ miles long. Allow about 1½ hours.

- Zig Zag Trail: 1 mile. Allow about 1 ½ hours.

EXPLORE (GEOLOGY) MAP: *Picture spread + 5 icons:*

1) ***Fossils found in Bishop's Quarry:***

Corals	Corals are marine animals which are often mistaken for plants.
Brachiopods	Brachiopods have shells and a stalk attaching them to the floor.
Trilobites	Trilobites were arthropods which are now extinct.
Crinoids	Crinoids, also called sea lilies, are actually marine animals

2) ***Limestone Pavement:***

As the Ice Age glaciers retreated, they exposed the limestone to rain, which washed into the cracks and eroded them into deep fissures or "grykes" between blocks of limestone known as "clints". This creates the limestone pavement, a rare and fragile feature which provides a home for many unusual and delicate plants.

3) ***Rock Contours:***

These layers of rock have been twisted and tilted by underground volcanic forces. As the sea wore away the cliff face, the layers, or "stratifications", have been revealed.

4) ***Copper Mineral Vein:***

This coloured stripe in the rock is copper ore, which has been mined here since the Bronze Age; you can go round the ancient mines and see how important copper once was.

5) ***Glacial Erratics:***

These rocks were carried here inside Ice Age glaciers and dropped as the ice melted; their unusual shapes have inspired local superstitions, and many have been given special names.

DISCOVER HISTORY

INTRO PAGE 1:

Discover Man's Place on the Great Orme

Today, the Great Orme is a Country Park, a place for recreation and relaxing. But people have lived and worked here for centuries. As you walk around, it is worth remembering the people who were here before you. So join our Country Park Warden as he reveals some of the secrets of man's impact on the peninsula…

Q1: The name "Great Orme" is believed to come from the Norse (Viking) word "orm" meaning

sea serpent □ ✔ dragon □

PAGE 2:

Viking sailors may have thought that the headland looked like the head of a great sea serpent in the mist, and are believed to have called the Orme "Horma Heva". But they were not the first people to come here…

Q2: Can you guess when the first visitors came to the Great Orme?

2,200 years ago □ 3,500 years ago □ 15,000 years ago □✔

PAGE 3:

The first inhabitants of the Great Orme were Stone Age hunter-gatherer tribes who came across from Europe over 15,000 years ago when Britain was still joined to the European mainland. They sheltered in the many caves around the Orme, and remains have been found of stone axes, flint scrapers and decorated animal bones.

Q3: Which metal do you think was used first to make tools and weapons?

iron □ bronze □✔ gold □

PAGE 4:

Bronze is a hard metal made from copper & tin, and the local copper deposits allowed an important mining community to flourish here around 3,500 years ago. Bronze was so important to early mankind that this period and its peoples are called the Bronze Age.

PAGE 5:

Around 800 years BC, Celtic tribes from Europe brought iron-working skills and built well-defended hill-forts and elaborate burial monuments. The hill fort of Pen Dinas contains the remains of over 60 roundhouses, and there are also many important Stone Age archaeological finds here on the Great Orme.

PAGE 6:
There is no evidence that the Romans settled here, but there are medieval settlements on the Great Orme which show a well-established community.

Cyngreawdr stood around St Tudno's church, which survives to the present day. You can see the remains of houses just below the summit, and clearly visible remains of the medieval "ridge and furrow" farming system.

PAGE 7:
Farming continues today at Pink Farm and Parc Farm. Much of the headland became a Country Park in 1980 and tourism is now the main activity of the Great Orme. Today, the Park attracts over 600,000 visitors a year.

Q4: When do you think the trend for sight-seeing and tourism started?

The 1770s □ The 1870s □✔ The 1970s □

PAGE 8:
In Victorian Britain, shorter working hours and the wealth created by the Industrial Revolution created a boom in leisure activities for all classes. From 1878, the rich could tour the Marine Drive toll road in their horse drawn carriages, and in 1902, the Tramway was completed, bringing crowds to the summit area, where an old telegraph station had been rebuilt as a hotel and is now a café.

PAGE 9:
The Great Orme was a vital part of the war effort during the Second World War: the Royal Artillery coastal gunnery school relocated from Shoeburyness in Essex to the west shore of the Great Orme in the summer of 1940 to avoid damage from enemy bombing. The shallow seas off the headland provided the ideal place for anchoring target ships.

PAGE 10:
You can explore the map to see where all these features are, or discover many more fascinating features of the Great Orme's history by **Taking a Tour** *(brochure pic) + pop-up text:*

The Historical Trails booklet can be bought at the visitor centre shop, and contains two circular walks from the summit of the Great Orme over a landscape rich in historical remains. Route A takes around 2 ½ hours, Route B takes 2 hours. They can be combined into one route of 7.4 miles. The track is steep and uneven in places, and walking boots are recommended.

EXPLORE (HISTORY) MAP: *Picture spread + 9 icons:*

1) ***A Stone Age burial chamber***

"Llety'r Filiast" (literally translated as "lair of the female greyhound") is a spectacular example of a Stone Age cromlech or burial chamber approximately 10 minutes walk from the visitor centre, near the Great Orme Ancient Mines.

2) ***Kendrick's Cave***

In 1880 local stone-mason Mr. Thomas Kendrick was clearing a cave in his garden to extend his workshop and discovered a number of bones, including a remarkable decorated horse's jaw. Further excavation has revealed the site as an important Stone Age burial site, and you can see the some of the remains in the Llandudno Museum.

3) ***The Great Orme Copper Mine***

Visit the Great Orme Ancient Mines, explore some of the Bronze Age tunnels, and discover how the Bronze Age miners lit fires deep underground to shatter the rock and work the seams of copper ore with primitive stone hammers and animal bones or even antlers as picks.

In the 17th century there was new demand for copper, and mining began again on a grand scale. You can see surface mines and spoil heaps between the summit and the halfway tram station, as the more modern miners first exploited easily accessible seams near the surface, and then deeper excavations underground.

4) ***Pen Dinas Iron Age hill-fort & rocking stone***

This would have been an ideal place for such a settlement, being protected on three sides by steep cliffs, with only one side needing the construction of a built double bank and ditch defence. Inside the fort there was a large community of over 60 roundhouses.

The 'Rocking Stone', is believed to be a way of trying people of certain crimes. The prisoner was stood upon the stone blindfolded – if it rocked they were innocent; if it stayed still they would have been thrown down the cliffs!

5) **Medieval Ridge and Furrow Agriculture**

Near St Tudno's church are extensive and easily visible remains of the medieval "ridge and furrow" agricultural system, where individual families ploughed strips of land in an open field system. The abandoning of growing crops in favour of grazing animals in later medieval times has preserved this feature.

6) ***Shipwrecks***

The Great Orme headland was a major navigational hazard to ships travelling to Liverpool and there have been many shipwrecks. On New Year's Day 1824, the

“Hornby” struck the rocks and was wrecked. A crew member called John Williams was thrown onto the cliffs, and scrambled to safety. He settled in the area and became a miner, telling his tale of survival, and the cave below the cliffs became known as Hornby cave. The Llandudno lifeboat was first launched in 1861 to rescue the many shipwrecked sailors of this period.

7) ***The Lighthouse***

The Great Orme lighthouse was built in 1862 by the Mersey Docks and Harbour Company to help guide shipping into the busy port of Liverpool. After closure in 1985 it was turned into a bed and breakfast hotel – but the original light was saved and is now here in the visitor centre.

8) ***The Tramway***

Great Britain's only remaining cable-operated street tramway was built in 1902, and runs from Victoria Station in Church Walks, Llandudno, to the Great Orme Summit. Passengers change tramcars, each named after a Welsh saint, at the Halfway Station.

9) ***Wartime Great Orme***

The gun and searchlight emplacements of the Royal Artillery coastal gunnery school can be explored on foot or viewed from above on the Marine Drive.

The summit complex was used as a top secret radar research station, and at the far western end of the headland, a “concrete road” is the old tank track to what was a three-storey building believed to have been a secret experimental radar station.

Around Bishop’s Quarry is an area known as the Hill of Names where it is said, the limestone rocks were originally arranged into the names of their regiments by the soldiers of the Royal Artillery School during the Second World War; people today continue this tradition by writing their own names.

DISCOVER WILDLIFE

INTRO PAGE 1:

Discover the Wildlife of the Great Orme

The Great Orme is a Country Park, a nationally important Site of Special Scientific Interest and has international status as a Special Area of Conservation. The headland offers a rich variation in habitats and some of the plants and animals you can see here are unique to the Great Orme.

PAGE 2:

You can walk **the Great Orme Nature Trail**. *(pic of brochure)* This circular walk around the headland, starting and finishing at the summit, is approximately 3 hours and 6.5 km/4 miles long, with a shorter route available. The track is steep and uneven in places, and walking boots are recommended.

Or you could go your own way and track down the wildflowers in our Wildflower Chart. *(pic of brochure)* The booklets are available from the visitor centre shop or can be downloaded from the Countryside pages of Conwy County Borough Council's website.

Now let's explore nature's year on the Great Orme and see just what makes the Great Orme so special…

EXPLORE (WILDLIFE):
MENU CHOICE OF 4 MAPS + ICONS, ONE FOR EACH SEASON:

SPRING ICON 1:

As nature wakes up after the winter, spring marks the arrival of goat kids on the Great Orme. Usually difficult to spot, watch out for them if you see a group of nanny goats (females), but please don't disturb them.

These are Kashmir goats, much prized for their wool, acquired from Queen Victoria's Royal herd by Major General Sir Savage Mostyn, and brought here in the 1900s; they have roamed wild ever since.

SPRING ICON 2:

Spring sees our biggest visitor numbers: but these visitors come all the way from Africa! Ring Ouzels, Tree Pipits and Whinchats stop here to "refuel" as they follow the spring migration north.

Swallows, Wheatears and Willow Warblers complete their migration here, settling on the Great Orme to breed each summer.

SPRING ICON 3:
Cotoneaster cambricus or the 'Great Orme berry' is found nowhere else but on limestone cliffs of the Great Orme and is Critically Endangered. It has small white flowers in spring and berries like miniature apples in the autumn. It is these berries that give the plant its Welsh name, Craig Afal (Rock Apple).

SPRING ICON 4:
Other rare wildflowers include spring cinquefoil, spring sandwort and hutchinsia. Thrift can be seen carpeting areas of grassland and sea cliffs towards the end of spring.

SPRING ICON 5:
On the northern side of the Great Orme are relics of the woodland that once grew on many parts of the headland. Plants like violets, wood sorrel, wood anemones and primroses now thrive under a canopy of bracken. This habitat provides an important refuge for woodland plants and the insects that rely on them.

SUMMER ICON 1:
During the summer months, spiked speedwell, common rockrose, hoary rockrose and the purple flowering wild thyme can be seen carpeting areas of limestone grassland.

SUMMER ICON 2:
The beautiful dark-red helleborine is a species of orchid which is endangered in Wales, so seeing it on the cliffs is a real treat. Keep an eye out for it in June and July.

SUMMER ICON 3:
The sea cliffs come alive as hundreds of kittiwakes, razorbills, cormorants, fulmars and guillemots arrive in spring to nest on narrow cliff ledges. Just follow the sound of their calls, but take care; cliffs can be dangerous!

SUMMER ICON 4:
Beginning to flower in late spring and continuing throughout the first 2 months of summer, rose pink thrift plants carpet the grasslands.

SUMMER ICON 5:
The grayling butterfly is a master of disguise. The sub-species found on the Great Orme is unique; it cannot be found anywhere else!

SUMMER ICON 6:
The silver-studded blue butterfly is small but beautiful, and the sub-species seen here is unique to the Great Orme.

SUMMER ICON 7:
Listen for the call of the Great Orme's resident stonechats among the gorse bushes: distinctive clicks that sound like stones tapped together, giving the bird its name.

AUTUMN ICON 1:
In the autumn you can see the billy goats clashing horns to compete for the attention of the nanny goats.

AUTUMN ICON 2:
Ring ouzels, tree pipits and whinchats can be seen feeding during their autumn migration as they stop to refuel on their way south to Africa for the winter.

AUTUMN ICON 3:
Now is the time to see the small red berries of the wild cotoneaster on the limestone cliffs.

AUTUMN ICON 4:
The woodlands are rich with mosses, fungi and lichens, and red campion flowers can be seen up to October.

WINTER ICON 1:
Winter sees the arrival of birds from Northern Europe and the Arctic such as the snow bunting, which searches for seeds on the generally snow-free ground as it over-winters here.

WINTER ICON 2:
Lady ferns and harts-tongue ferns can be seen in the cracks of the limestone pavement.

WINTER ICON 3:
Hardy domestic sheep will be out in all weathers, and so will our Country Park Wardens; the heathlands are cut every winter to encourage new seedlings and fresh ground cover.

EXPLORE (ATTRACTIONS) MAP:
MENU CHOICE OF "THINGS TO DO" OR "WHERE TO EAT" ICONS:

ATTRACTIONS ICON 1:

Ski Slope and Snowboard Centre
With the longest toboggan run in Britain and a scenic PermaSnow ski slope, the centre offers skiing, snowboarding, toboggan & sno-tubing with a fully licensed Alpine Lodge Bar & Restaurant.

Open 10am - 10pm
http://jnllandudno.co.uk
01492 874707

ATTRACTIONS ICON 2:

The Great Orme Ancient Mine
The copper mines date back to the Bronze Age, and are one of the most astounding archaeological discoveries of recent times.

Open daily mid-March to end of October
Open 10am - last tour 4:30pm
http://www.greatormemines.info/
01492 870447

ATTRACTIONS ICON 3:

The Cable Car
Running from Happy Valley to the Great Orme Summit, the Cable Car reveals many magnificent views, and at one mile from end to end, is the longest aerial cabin lift in the United Kingdom.

http://www.greatorme.org.uk/cablecar.html

ATTRACTIONS ICON 4:

The Tramway
Great Britain's only remaining cable operated street tramway runs from Victoria Station in Church Walks, Llandudno, to the Great Orme Summit, with passengers changing cars at the Halfway Station.

Open mid-March to early November 10am - 5.45pm daily
http://www.greatormetramway.co.uk/
01492 577877

ATTRACTIONS ICON 5:

The Olde Victorian Picture House
Create your unforgettable holiday picture with costumes, a studio with a variety of themes and modern technology used to re-create, perfectly, the atmosphere of days gone by. Part of the Summit Complex.

Open daily 11am - 5pm from mid March to early November
http://www.theoldevictorianpicturehouse.co.uk/
01492 860870

ATTRACTIONS ICON 6:

Rocky Pines Adventure Golf
Part of the Summit Complex, Rocky Pines Adventure Golf is an 18-hole course providing spectacular views of the dramatic North Wales coast.

Open daily 11am - 5pm from mid March to early November
http://www.theoldevictorianpicturehouse.co.uk/golf.htm
01492 860870

ATTRACTIONS ICON 7:

Great Orme Family Golf
The only full sized 9-hole golf course in North Wales, with spectacular views and fun for the whole family.

Open daily 10am-dusk
www.Ormegolf.blogspot.com
07990 517020

ATTRACTIONS ICON 8:

Marine Drive
A 4 ½ mile scenic drive around the Great Orme headland, from Llandudno's North Shore to Llandudno's West Shore, with wonderful views in all directions.

£2.50 per car (which also covers parking at the summit of the Great Orme)

WHERE TO EAT ICON 1:

The Summit Complex
The Summit Complex consists of a café / restaurant and boxing themed bar with breath-taking views.

WHERE TO EAT ICON 2:

Rest and be Thankful Cafe
As well as serving rolls, cakes, drinks, ice creams and many other delightful treats, the Cafe also acts as an information centre for the Great Orme and maintains a large range of photograph albums and specialist publications.

http://www.restandbethankful.net/

WHERE TO EAT ICON 3:

Alpine Lodge
The Alpine Lodge bar and restaurant provides panoramic views of the main ski slope, toboggan track, Alpine Adventure Golf course and Llandudno bay.

http://jnllandudno.co.uk
01492 874707

WHERE TO EAT ICON 4:

Parisella's Cafeteria
Set in Happy Valley gardens, Parisella's Cafeteria provides drinks, cakes, hot meals and Parisella's own ice cream.

www.parisellasicecream.co.uk
01492 592448

WHERE TO EAT ICON 5:

Haulfre Gardens Tea Rooms
A traditional tea room set in the beautiful and relaxing location of Haulfre Gardens, serving homemade cakes, scones and bara brith. A sheltered sun-trap, with panoramic views across Llandudno including the West Shore and Conwy Estuary.

Open daily 10am-7pm
01492 876731